

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2004

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

- 4.** Which one of the following components of blood is involved in the clotting after injury?
- A. Plasma
 - B. White blood cells
 - C. Red blood cells

D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C] [D] 24. [A] [B] [C] [D] 34. [A] [B] [C] [D]

11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

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1. After incubation period the chick comes out of the shell by
 - A. The mother cracking the shell open using the beak
 - B. Cracking the shell open using its beak
 - C. Growing bigger thus forcing the shell to break
 - D. The shell cracking on its own
2. Some curative drugs are also called
 - A. Antibiotics
 - B. Vaccine
 - C. Stimulants
 - D. Painkillers
3. A jar for measuring rainfall should be
 - A. Narrow
 - B. Wide
 - C. Short
 - D. Long
4. Which one of the following statements about adaptations in animals is NOT CORRECT?
 - A. Cheeters have sharp, curved claws to help them catch their prey
 - B. Eagles have strong, curved beaks to help them cut and tear fresh
 - C. Crocodile have saw-like teeth to help them grip their firmly
 - D. Snakes have fang to hold their prey firmly
5. Which one of the following is formed first
 - A. Foetus
 - B. Embryo
 - C. Zygote
 - D. Sex cell
6. Which one of the following does not increase the strength of an electromagnetic?
 - A. Number of coils
 - B. Amount of current
 - C. Size of soft iron
 - D. Length of connecting wires
7. A change in the force of gravity on an objects will change its
 - A. Mass
 - B. Weight
 - C. Volume
 - D. Density
8. Which one of the following machine is the effort equal to the load and the effort distance equal to the load distance?
 - A. Incline plane
 - B. Wheel and axle
 - C. Lever
 - D. Single fixed pulley
9. The force that act on given area is called

- A. Weight
 - B. Gravity
 - C. Pressure
 - D. Friction
10. Which one of the following properties of clay soil would greatly contribute to flooding in an area
- A. High capillarity
 - B. Poor drainage
 - C. Fine texture
 - D. Stickiness
11. The main reason why sandy soil is mixed with cement in building is because it?
- A. Drains well
 - B. Has large air space
 - C. Has large particles
 - D. Mixes easily
12. When choosing a method of separating a mixture of liquid and solid it is important to consider the
- A. Density of the liquid
 - B. Size of the particles of the solid
 - C. Density of the solid
 - D. Solubility of the solid
13. Which one of the following is not definite for liquids?
- A. Shape
 - B. Volume
 - C. Mass
 - D. Density
14. In feeding relationships , mice belongs to the group as
- A. Vultures
 - B. Snakes
 - C. Antelopes
 - D. Lions
15. The table below shows parts of the human digestive system and their functions

Which one of the following option represents X and Y?

X	Y
A. Oesophagus	passage of food
B. Large intestine	digestion of food
C. Rectum	storage of undigested food
D. Mouth	taking food

16. The following were reasons given by pupils to explain why smoke rose. It?
- i) Has hot air in it
 - ii) Is less denser

- iii) Has particles suspended in air
- iv) Is cloudy

Which two reason are correct

- A. (i) and (ii)
 - B. (iii) and (iv)
 - C. (ii) and (iii)
 - D. (i) and (iv)
17. A one meter long piece of wood floats on water while a one centimeter long piece of iron rod sinks
This is because of difference in
- A. Length
 - B. Material
 - C. Weight
 - D. Shape
18. The diagram below represents a fish rod with points labeled P, Q, R and S
Which in use the fulcrum is at point
- A. P
 - B. Q
 - C. R
 - D. S
19. The table below shows observation made when four soil samples P, Q, R and S were mixed with water and then rolled out to make ribbons

Soil sample	characteristic
P	held its shape
Q	cracked
R	could not be rolled
S	could not hold its shape

Which one of the sample has the lowest capillarity?

- A. P
 - B. Q
 - C. R
 - D. S
20. Which one of the following drugs causes drowsiness?
- A. Cocaine
 - B. Heroine
 - C. Tobacco
 - D. Khats
21. The diagram below represents an improvised liquid thermometer

Which one of the following improvement would make the instrument measure smaller changes in temperature?

- A. Making the straw narrower
- B. Colour the water
- C. Using a plastic
- D. Using a larger bottle

22. The chart below shows a simple classification of small animals

Which animal is represented by X and Y

- | X | Y |
|--------------|--------|
| A. Butterfly | fleas |
| B. Wasp | crabs |
| C. Cockroach | tick |
| D. Mosquito | spider |

23. Which one of the following practices help to conserve soil by preventing soil erosion

- A. Crop rotation
- B. Use of manure
- C. Mulching
- D. Planting cover crops

24. Which one of the following pairs of processes involves absorption of heat from the surroundings?

- A. Condensation and evaporation
- B. Freezing and condensation
- C. Melting and freezing
- D. Evaporation and melting

25. The diagram below shows the beak of a certain bird

The bird most likely feeds on

- A. Insects
- B. Nuts
- C. Grains
- D. Flesh

26. If the size of hole camera is increased, the image formed is

- A. ENLARGED
- B. BLURRED
- C. UPRIGHT
- D. NOT REAL

27. Which one of the following sources of energy is non- renewable?

- A. Kerosene
- B. Firewood
- C. Solar
- D. Biogas

28. A stone at the bottom of a pond appears nearer the surface due to
- Refraction
 - Dispersion
 - Diffraction
 - Reflection
29. Which one of the following shows the correct order of parts in human eye through which light passes?
- Cornea -> pupil-> lens-> retina
 - Pupil-> lens-> cornea->retina
 - Pupil-> cornea->lens->retina
 - Cornea-> lens-> pupil-> retina
30. Which one of the following parts of passion plants is modified for climbing?
- Stem
 - Branch
 - Leaf
 - Bud
31. Which one of the following waste materials is not a product of excretion?
- Sweat
 - Urine
 - Carbon dioxide
 - Faeces
32. Which one of the following describes cross-pollination?
- The transfer of pollen grain from the anthers to the stigma of the same flower
 - The transfer of pollen grain from the anthers to the stigma of different flower on plants of different kind
 - The transfer of pollen grain from anthers to the stigma of another flower ON THE SAME PLANT
 - The transfer of pollen grains from the anthers to the stigma of different flowers on plants of the same kind
33. The chart below represents approximate percentage of the component of air

Which portion represents the gas that is necessary for burning?

- P
 - Q
 - R
 - S
34. Which one of the following activities is correct when comparing the strength of an acid in the fruit juices? To ash solution add
- Indicator and then fruit juice drop by drop
 - Fruit juice and then indicator drop by drop
 - Equal amount of fruit juice and then indicator drop by drop
 - Equal amount of indicator and fruit juice

35. The diagram below represents a set up that is used to demonstrate a certain property of air
The demonstration shows that air
- A. Is dissolved in water
 - B. Has weight
 - C. Occupies spaces
 - D. Exerts pressure
36. Which one of the following statements about wind vane is not true
- A. The tail should be larger than the arrow head
 - B. It should be placed in an open place
 - C. The arrow points the direction in which the wind is blowing
 - D. Its compass points should correspond to the geographical compass points
37. In the diagram below P, Q, and S represents different layers of materials in water filter

Which material are represented by P, Q, R and S

38. Which one of the following materials cannot be recycled?
- A. Rubber tyres
 - B. Glass bottles
 - C. Candle wax
 - D. Saw dust
39. Which one of the following pairs of machines consists of levers only
- A. Plies and wedge
 - B. Crow bar and nutcrackers
 - C. Screw and scissors
 - D. Bottle opener and knife
40. Clothes a clothe line took to long to dry. The weather was most likely
- A. Dry and calm
 - B. Cold and windy
 - C. Dry and cold
 - D. Cold and calm
41. Standard five pupils observed a flower with brightly coloured petals in their compound.
Which one of the following likely to be the other characteristic of that flower?
- A. Feathery stigma and few pollen grains
 - B. Large pollen grains and feathery stigma
 - C. Sticky stigma and few pollen grains
 - D. Light pollen grains and sticky stigma
42. Which one of the following statements about blood circulation is true?
- A. Aorta receives blood from the left auricle
 - B. Oxygenated blood from the lungs is received by aorta
 - C. Pulmonary vein takes blood to lungs
 - D. Pulmonary artery receives deoxygenated blood from the right ventricle
43. Which one of the following is part of embryo of the seed
- A. Micropyle

- B. Endosperm
 - C. Radicle
 - D. Cotyledon
44. Which one of the following is not important to consider when taking medicine?
- A. The dose
 - B. The date of manufacturer
 - C. Duration to complete the dose
 - D. How to store the medicine
45. Which one of the following characteristic is true for both plants and animals
- A. Move, use carbon dioxide
 - B. Make their own food, grow
 - C. Reproduce, transpire
 - D. Excrete, respond to changes in the environment
46. Which one of the following practices is used for controlling pollution in soil, water and air?
- A. Avoid dumping can and plastic materials
 - B. Treating chemical waste from factories
 - C. Burning rubbish in metallic or concrete structures
 - D. Avoid excess use of fertilizers
47. In which of the following cases is friction both an advantage and disadvantage
- A. Dragging a heavy box along the floor
 - B. Moving parts of machines
 - C. Climbing up a hill
 - D. Drawing water from a well using a rope
48. Coins were arranged in pie as shown in the diagram below
When the coin on the bottom was knocked out suddenly using a ruler it moved out leaving the pie intact
The pie remained intact because of
- A. Friction
 - B. Gravity
 - C. Inertia
 - D. Weight
49. An inclined plane makes work easier by
- A. Changing direction of effort
 - B. Increasing effort distance
 - C. Decrease load distance
 - D. Decrease the load
50. Which one of the following is not important when comparing solubility of solids in liquid?
- A. Amount of the solid
 - B. Size of containers
 - C. Temperature of liquid
 - D. Amount of liquid